

Muhammad Azam Afridi

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Portfolio
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About Me

Erasmus Mundus Master's Scholar in Linguistic Data Science (EMLDS) and Gold Medalist in Artificial Intelligence, with professional experience as an AI Engineer. My research interests include large language models, particularly LLM agents, prompting and reasoning techniques, and methods for building more capable and effective AI systems.

Education

Erasmus Mundus Joint MSc in Linguistic Data Science (EMLDS) Sep 2025 – Present
Università Cattolica del Sacro Cuore (Italy) · University of Zaragoza (Spain) · Universidade NOVA de Lisboa (Portugal)

- Awarded a fully funded Erasmus Mundus master's scholarship.
- Weighted average: **29.46/30**; earned **30 e lode** in three of four completed examinations.
- Coursework: Machine Learning & Deep Learning, Language Models, NLP, Speech Processing

Bachelor of Science in Artificial Intelligence Sep 2020 – Jun 2024
GIK Institute of Engineering Sciences and Technology (GIKI) Topi, Pakistan

- GPA: 3.75/4.0 — **Ranked 1st of 55 (Gold Medalist)**
- Thesis: LLM-Powered Multilingual Video Translation and Multimodal Question-Answering System.

Work Experience

AI Engineer Mar 2025 – Sep 2025
Tkxel Lahore, Pakistan

- Built a multi-agent orchestration system for Insphere, the firm's enterprise GenAI platform, coordinating specialized sub-agents for multi-step reasoning and task delegation.
- Designed the end-to-end Agentic RAG architecture (LangChain, LangGraph), cutting response latency by **15%** and improving retrieval accuracy by **7%** over baseline.
- Developed backend services and ingestion pipelines using FastAPI, PostgreSQL, and Redis; assessed prospective clients' requirements during discovery calls and recommended AI solution opportunities.

AI Engineer Jun 2023 – Mar 2025
Ninety Nine Technologies Islamabad, Pakistan

- Built a multimodal RAG system over company documents stored in Google Drive, automatically updating the vector database when source documents changed. Integrated text and vision embedding models with LLMs and VLMs to process and retrieve information from text, images, audio, and video.
- Re-engineered sequential product scraping into concurrent multi-scraper execution, increasing data collection throughput by **10×**.

Research Assistant Aug 2023 – Jun 2024
GIK Institute of Engineering Sciences and Technology Topi, Pakistan

- Conducted research on multilingual video translation and video-based question answering using speech-to-text, text-to-speech, machine translation, vision-language models, and OCR under the supervision of Dr. Khurram Khan Jadoon.

Research Experience

POS-Signature Demonstration Retrieval for In-Context Dependency Parsing | Paper | Code
***Muhammad Azam Afridi**, Rajab Ali, Marco Passarotti*

- Designed a POS-signature retrieval method that combines part-of-speech statistics with length filtering to select relevant few-shot parsing examples.
- Benchmarked six prompting strategies across two LLMs on the full 2,077-sentence English-EWT test set, with statistically validated improvements in LAS over semantic retrieval of 1.9 percentage points for gpt-oss-120b and 2.7 for Qwen2.5-72B-Instruct.

Selected Projects

Vision-Language Robot Grounding (2nd Runner-Up – ZHAW Challenge, Hack Apertus 2026) | Link

- Built a vision-language robot planning pipeline using SAM 3, Depth Anything V2, and Apertus 1.5 70B, achieving 63.0/65 automatic points.
- Engineered three-agent planning with JSON-schema constrained decoding and deterministic validation, achieving scores of 62.5–64.1, compared with 58.9–60.5 for a single-call LLM approach.

Parameter-Efficient Fine-Tuning of Flan-T5 for Code Intelligence | Link

- Fine-tuned **Flan-T5-base** (250M) using LoRA for Python code generation and text-to-SQL on the MBPP and SPIDER benchmarks, training only **1.4%** of model parameters.
- Built preprocessing, training, and evaluation pipelines with schema-enriched prompting and task-specific metrics including pass@1, AST parse success rate, exact match, ROUGE-L, and BLEU-2.
- Improved SPIDER relaxed exact match from 0% to 2.53% and ROUGE-L from 10.38 to 48.57.
- Created and published processed versions of the MBPP and SPIDER datasets on Hugging Face, making the project data publicly accessible and reusable. | Dataset 1, Dataset 2

Agentic RAG System for Document Intelligence | Link

- Built an end-to-end platform for document question answering, summarization, and workflow automation using tool-calling LLM agents and semantic retrieval.
- Developed asynchronous ingestion pipelines integrating Google Drive, Gmail, and AWS S3, with Celery for background processing and Streamlit for the application interface.

Multilingual Video Translation and Multimodal Question Answering System | Link

- Built an end-to-end system for multilingual video translation, combining Whisper-based speech recognition, machine translation, and text-to-speech to convert spoken video content into a target language.
- Developed a RAG-based chatbot that retrieves information from uploaded videos and answers context-specific questions using LLMs, embeddings, and vector databases such as FAISS/ChromaDB.
- Integrated multimodal processing by extracting video frames and textual information to enrich the retrieval pipeline, enabling question answering over both spoken and visual content.

Certifications

Specialization in Deep Learning (Coursera) | Link
Certification of Python for Machine Learning & Data Science Masterclass (Udemy) | Link
Specialization in Machine Learning (Coursera) | Link

Scholarships and awards

Fully funded Erasmus Mundus master's scholarship | EMLDS	Sep 2025
Faculty Gold Medal | GIK Institute	Jun 2024
Academic Honors — six-time recipient | GIK Institute	May 2024
Best AI Project Competition Winner | PAF-IAST	May 2024
Best CS Project Competition Winner | NaSCon	Mar 2024

VOLUNTEERING

Executive Admin Events | Team Techno Feb 2022 – Jun 2024

- Organized four technical workshops on Python and machine learning, and a coding competition, fostering skill development and collaboration among participants.

Event Coordinator | GIKI Sports Society Feb 2021 – Jun 2024

- Coordinated three intramural sports competitions, featuring football and basketball, and led a national-level sports event, enhancing athletic participation and teamwork.

IT Team Lead | Project Topi Sep 2021 – Jun 2024

- Developed a web application to connect the GIK community with resources and organized digital literacy workshops for the local community.

Technical Skills

- **Certifications**: Deep Learning Specialization (Coursera); Machine Learning Specialization (Coursera); Python for ML (Udemy).
- **Programming**: Python, JavaScript.
- **AI and machine learning**: PyTorch, TensorFlow, Hugging Face, PEFT/LoRA, LangChain, LangGraph; agentic RAG, multi-agent systems, language models, vision-language models, object detection, OCR.
- **Backend and data**: FastAPI, Django, PostgreSQL, pgvector, Redis, Celery.
- **Cloud and development**: AWS, Azure, Docker, Git.